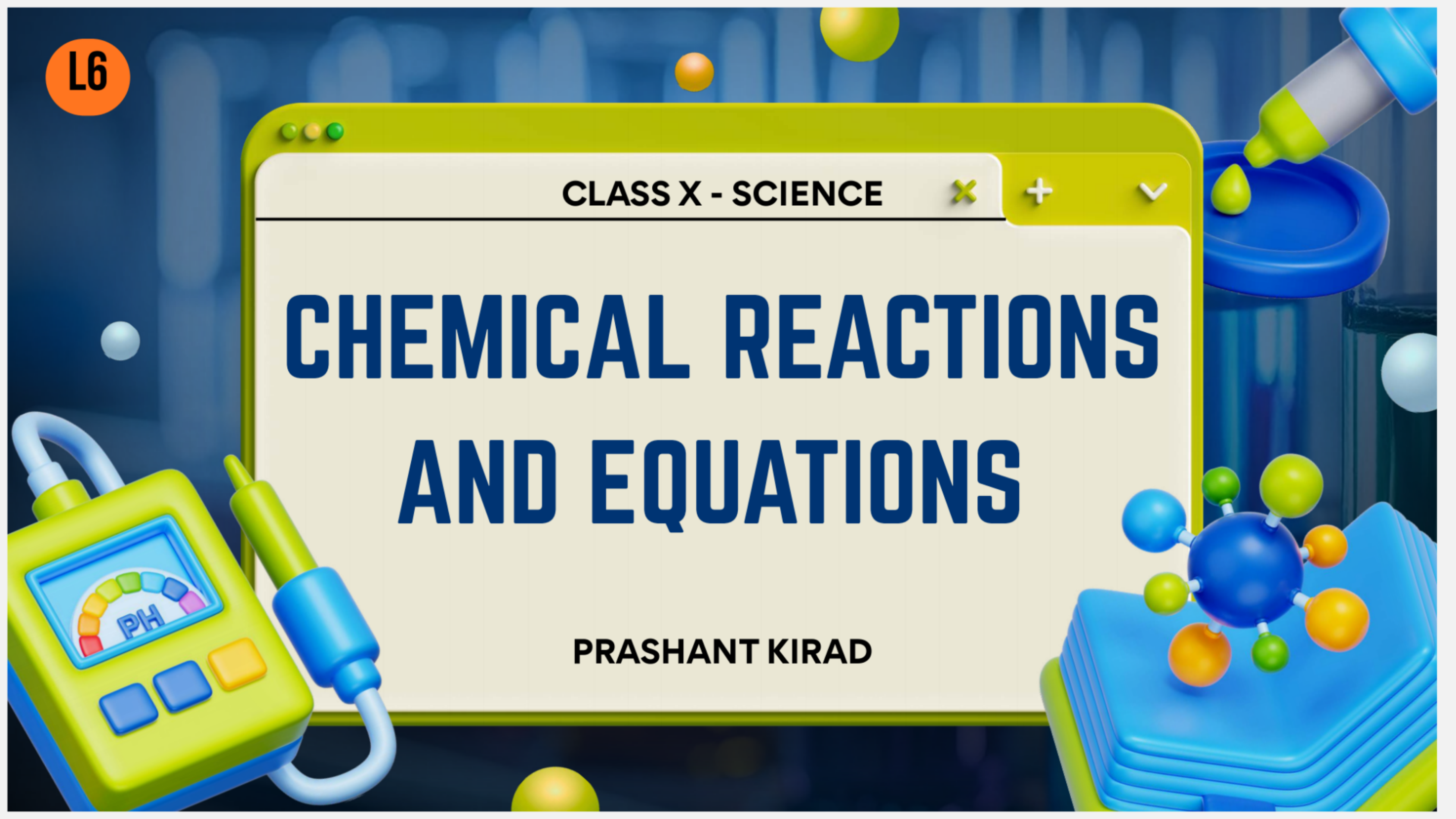


L6

CLASS X - SCIENCE

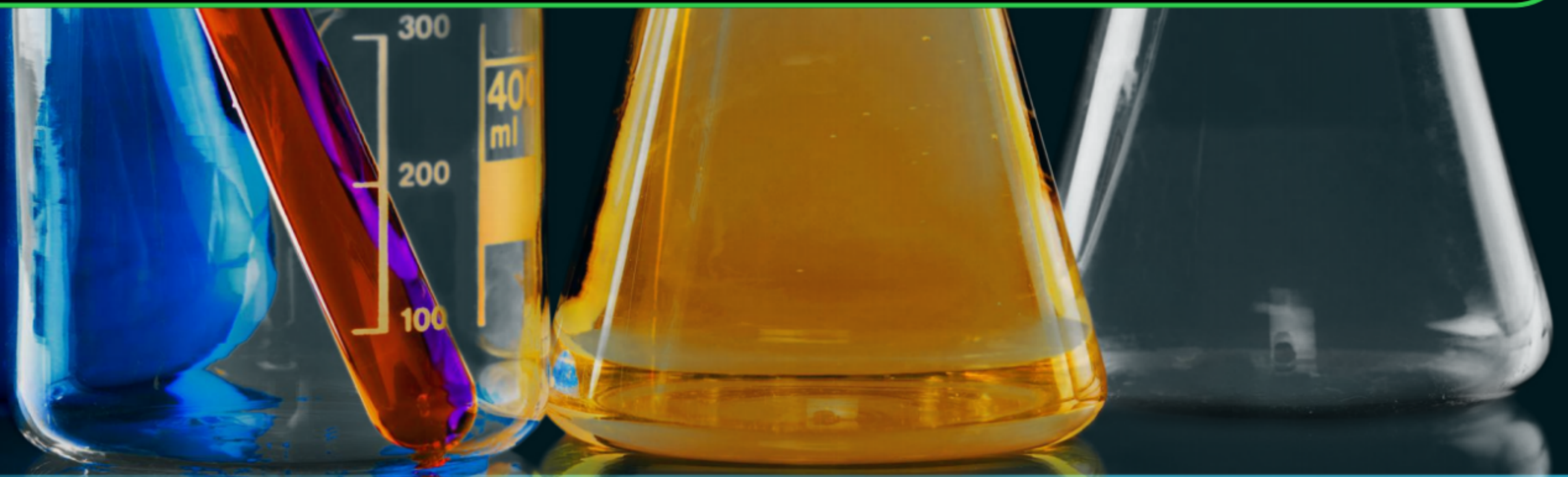
CHEMICAL REACTIONS AND EQUATIONS

PRASHANT KIRAD





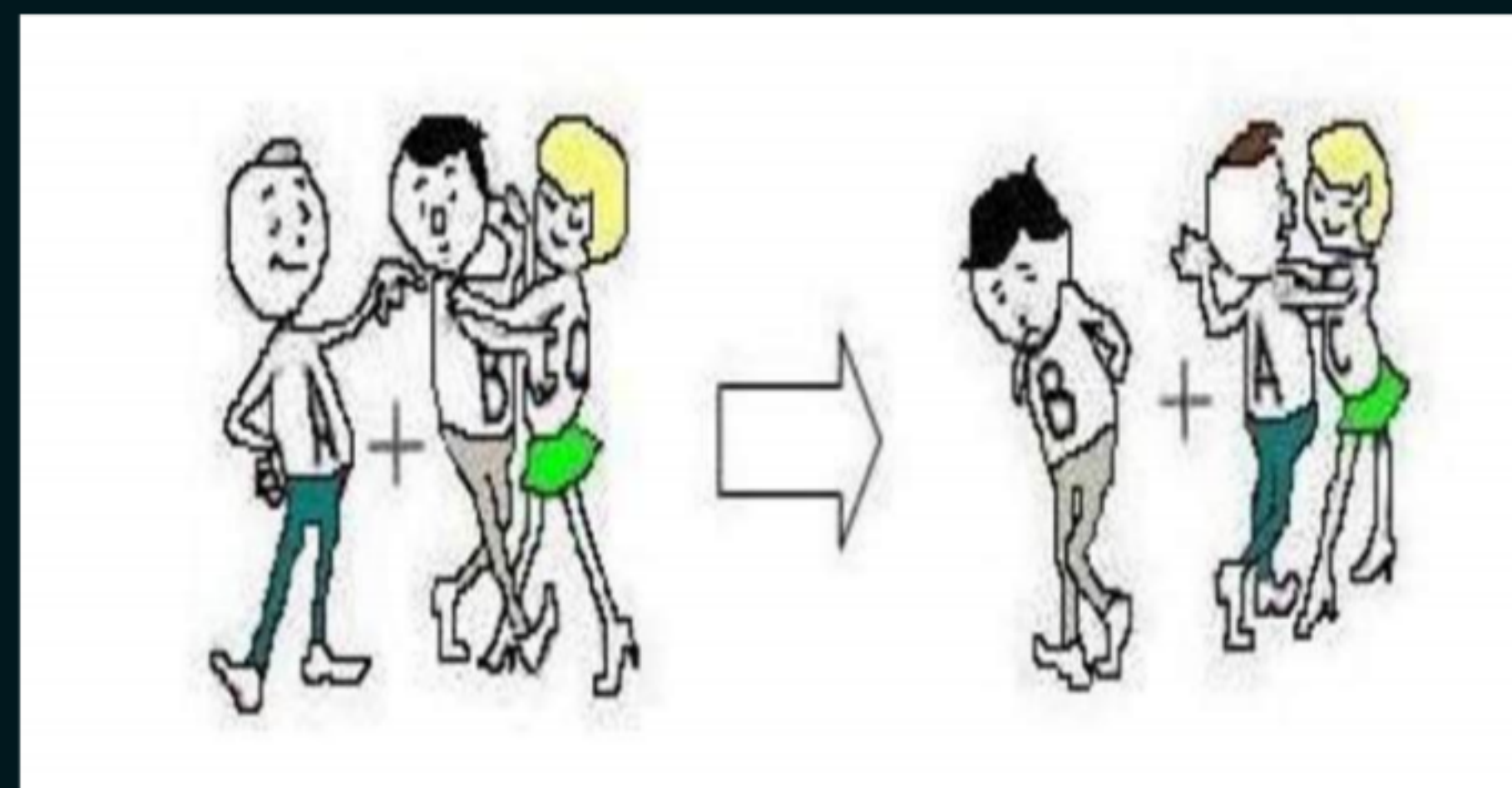
Displacement reaction and its types

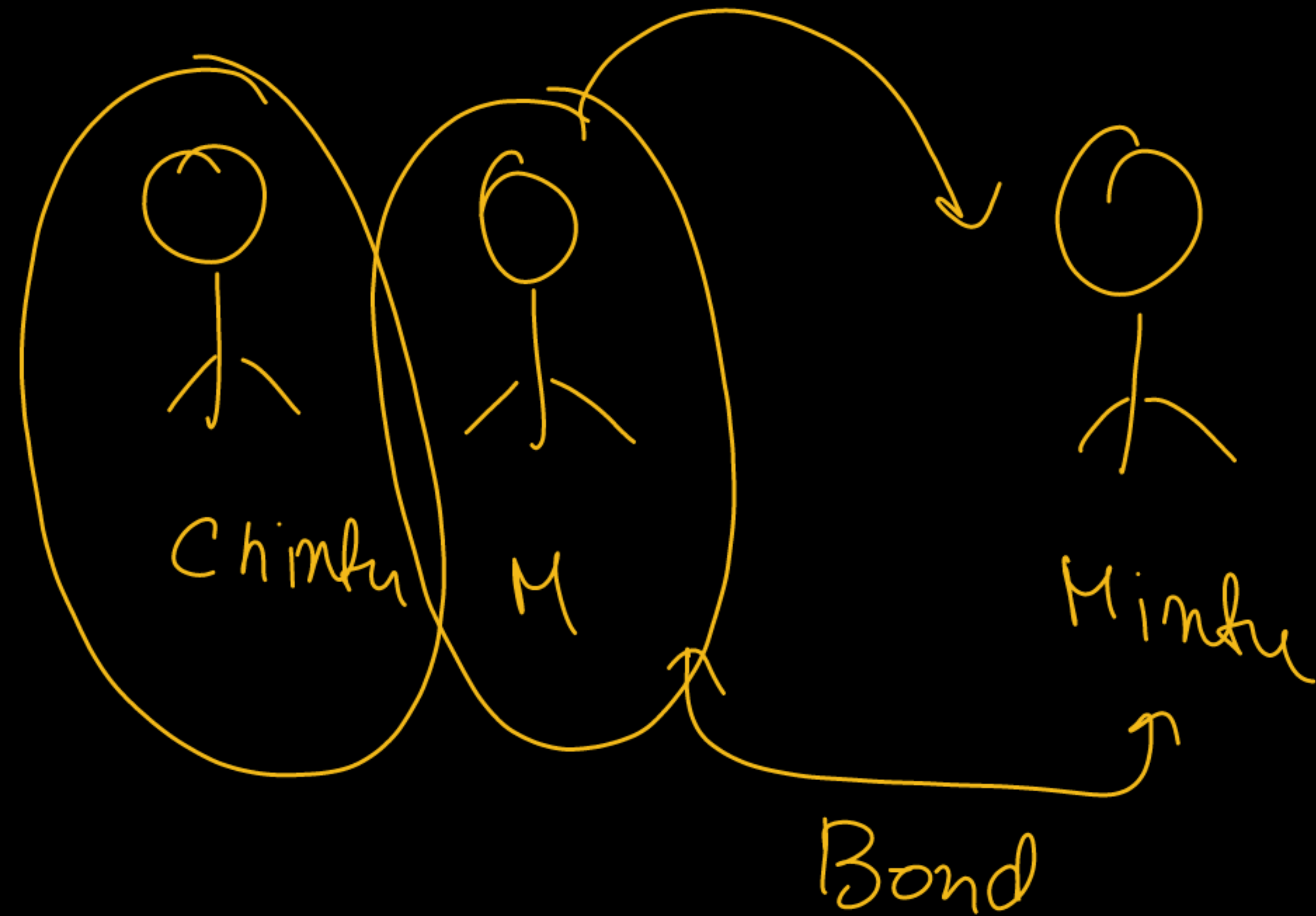


Displacement Reaction

A displacement reaction is a type of chemical reaction in which a more reactive element replaces a less reactive element from its compound.

★ General Formula:





Reactivity series

K	Potassium	Most reactive
Na	Sodium	
Ca	Calcium	
Mg	Magnesium	
Al	Aluminium	
Zn	Zinc	Reactivity decreases
Fe	Iron	
Pb	Lead	
H	Hydrogen	
Cu	Copper	
Hg	Mercury	
Ag	Silver	
Au	Gold	Least reactive

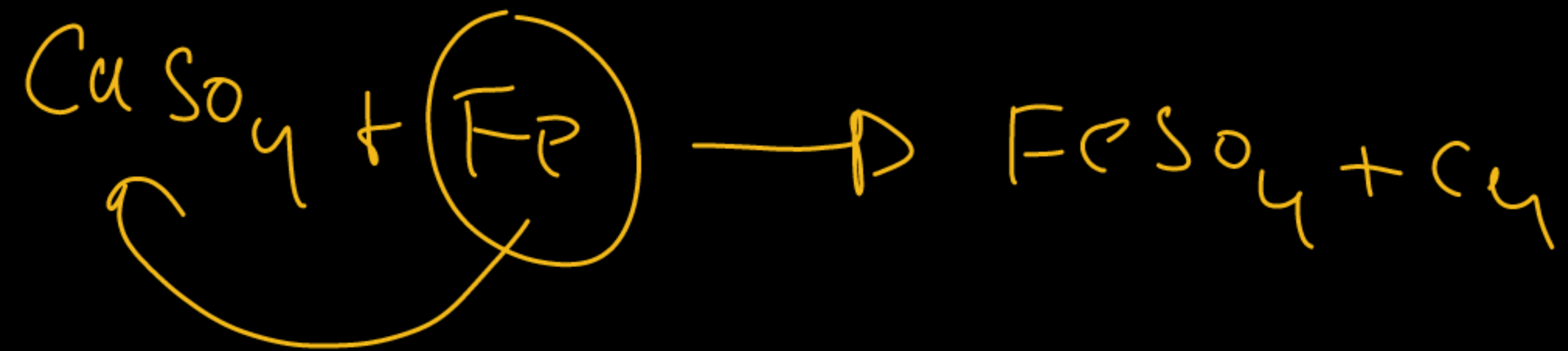
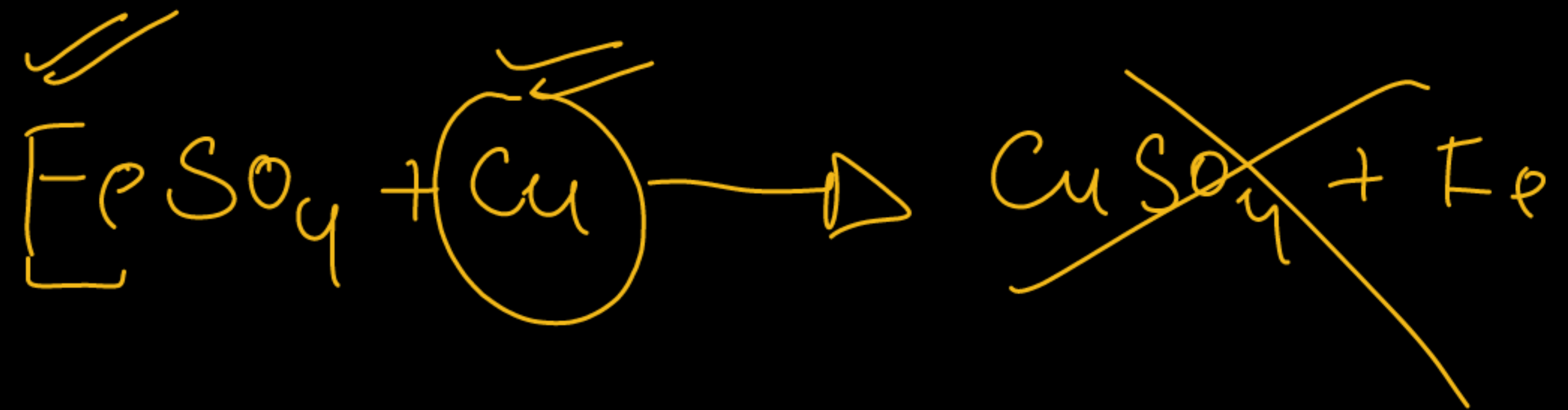
Trick to remember!

Metal-Metal Displacement Reaction



=



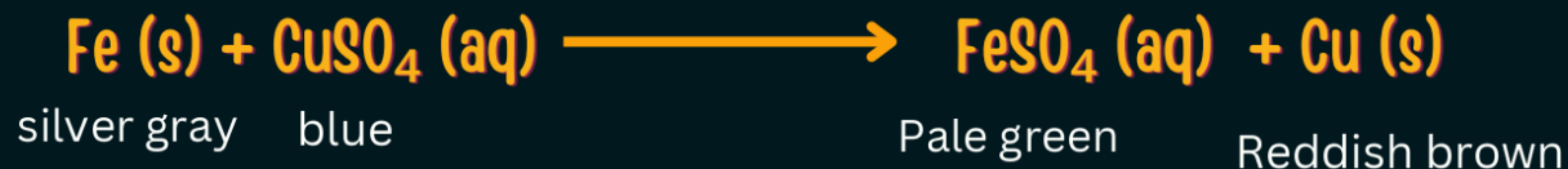


Metal-Metal Displacement Reaction



Why does the iron nail become brownish in colour and the blue colour of copper sulphate solution fades?

The following chemical reaction takes place in this Activity-



When iron is added to a copper sulphate solution, it displaces copper metal as iron is more reactive than copper.

NCERT ACTIVITY

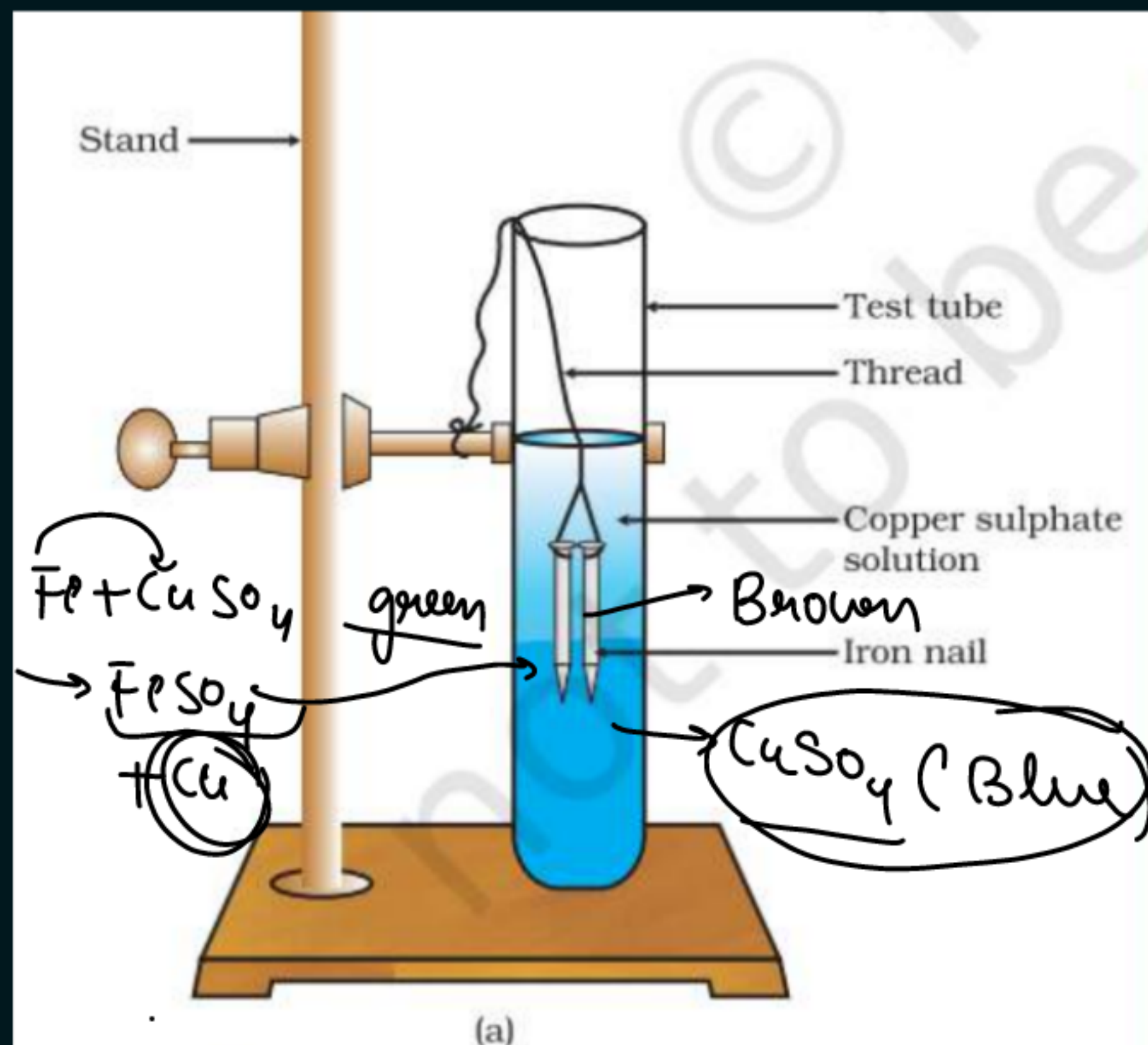


Figure 1.8
(a) Iron nails dipped in copper sulphate solution

Activity 1.9

- Take three iron nails and clean them by rubbing with sand paper.
- Take two test tubes marked as (A) and (B). In each test tube, take about 10 mL copper sulphate solution.
- Tie two iron nails with a thread and immerse them carefully in the copper sulphate solution in test tube B for about 20 minutes [Fig. 1.8 (a)]. Keep one iron nail aside for comparison.
- After 20 minutes, take out the iron nails from the copper sulphate solution.
- Compare the intensity of the blue colour of copper sulphate solutions in test tubes (A) and (B) [Fig. 1.8 (b)].
- Also, compare the colour of the iron nails dipped in the copper sulphate solution with the one kept aside [Fig. 1.8 (b)].

NCERT ACTIVITY

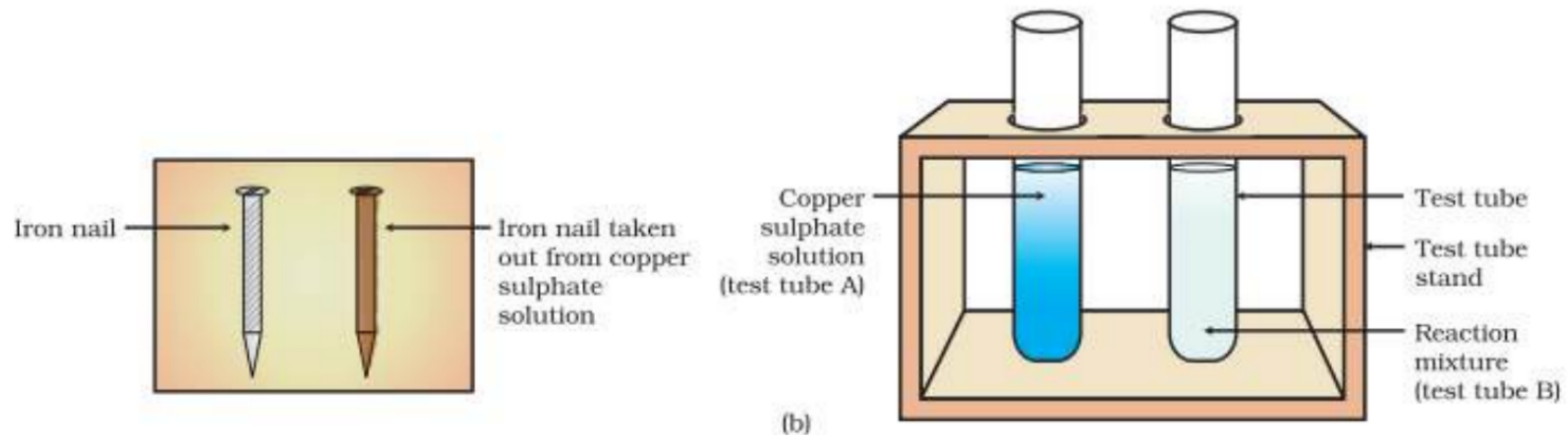


Figure 1.8 (b) Iron nails and copper sulphate solutions compared before and after the experiment

PK Points!

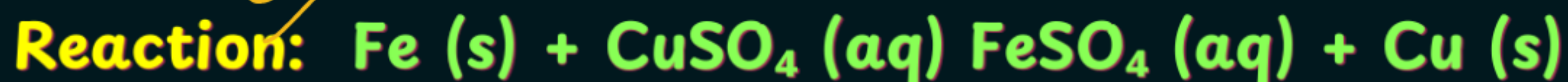


Observation:

- The blue colour of copper sulphate fades in test tube B. → green FeSO_4
- A reddish-brown deposit forms on the iron nails in test tube B.
- The iron nail kept aside remains shiny and unchanged.
- Iron is more reactive than copper. → Cu

Conclusion: Iron displaces copper from copper sulphate solution because iron is more reactive than copper.

This is a displacement reaction.



NCERT ACTIVITY



Activity 1.3

- Take a few zinc granules in a conical flask or a test tube.
- Add dilute hydrochloric acid or sulphuric acid to this (Fig. 1.2).
- CAUTION:** Handle the acid with care.
- Do you observe anything happening around the zinc granules?
- Touch the conical flask or test tube. Is there any change in its temperature?

Observations:

END

- Bubbling/Fizzing seen around zinc granules (gas is evolved).
- Temperature of the flask increases (feels warm on touching).
- The zinc granules slowly dissolve.

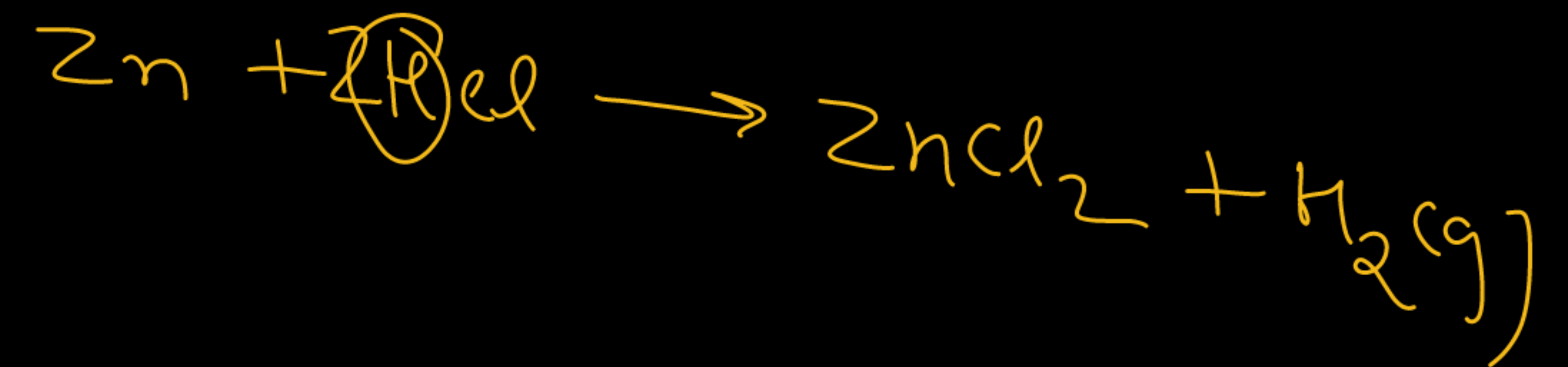
Chemical Equations:

- Using Hydrochloric Acid (HCl): $\text{Zn} + 2\text{HCl} \rightarrow \text{ZnCl}_2 + \text{H}_2 \uparrow$
- Using Sulphuric Acid (H_2SO_4): $\text{Zn} + \text{H}_2\text{SO}_4 \rightarrow \text{ZnSO}_4 + \text{H}_2 \uparrow$

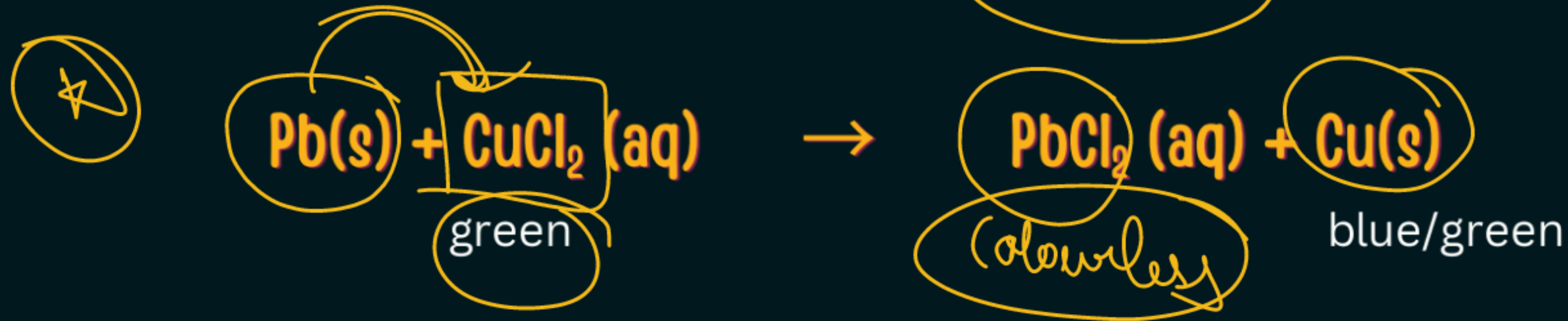
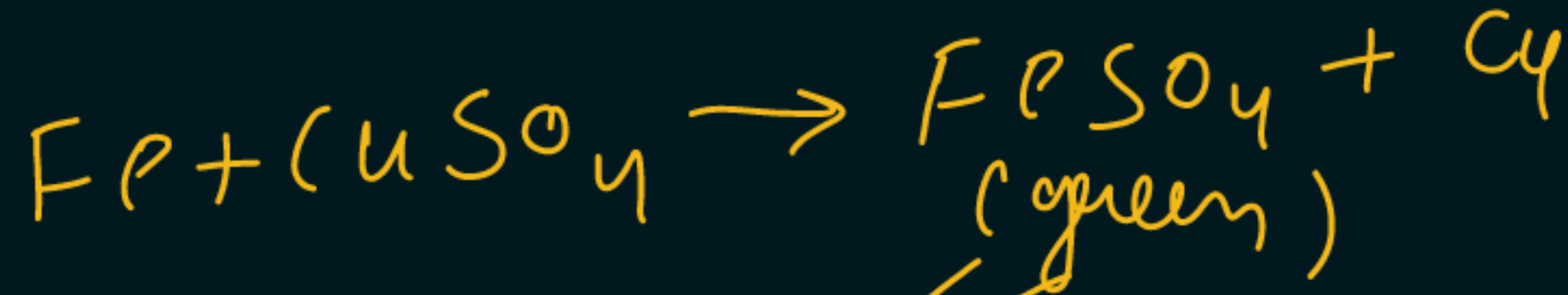
Important Notes:

- Hydrogen gas** is confirmed by bringing a burning matchstick near the mouth of the test tube – it produces a 'pop' sound.
- Heat released indicates the reaction is **exothermic**.





Examples -



Zinc and lead are more reactive elements than copper. They displace copper from its compounds

LET'S TEST YOUR MEMORY

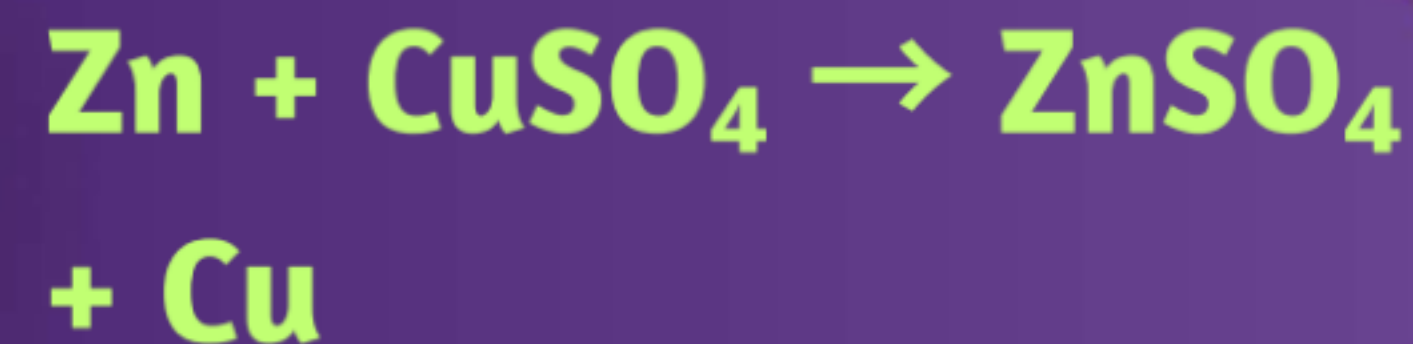


Q. Which of the following is a displacement reaction?
(CBSE 2020)

A



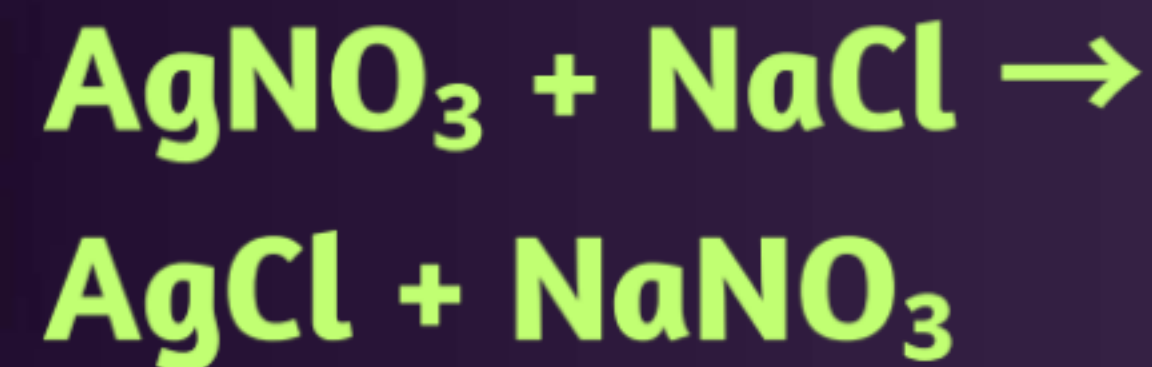
B



C



D



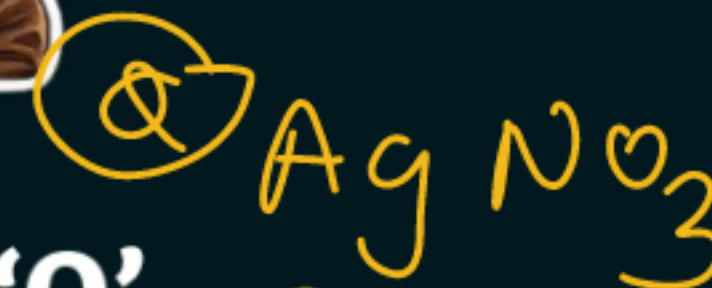
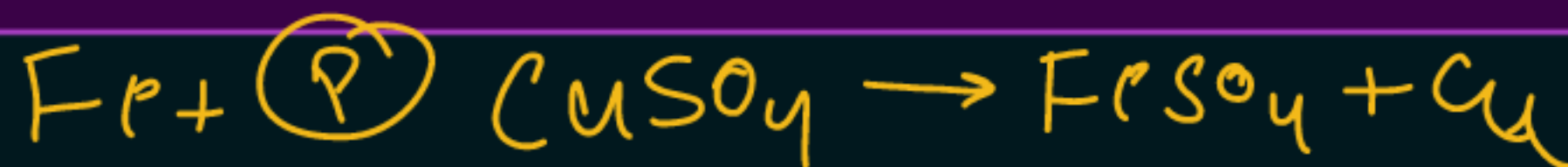
B



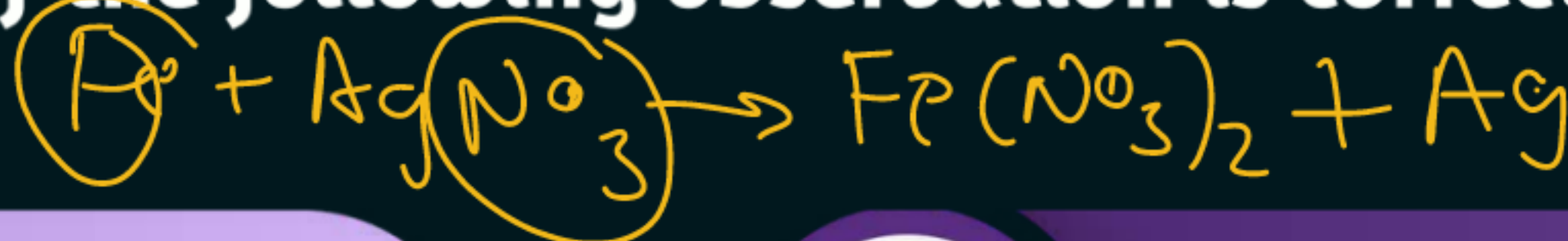
LET'S TEST YOUR MEMORY



2027



Q. An iron nail is added to each of the two test tubes 'P' and 'Q' containing aqueous copper(II) sulphate, and aqueous silver nitrate, respectively. Which of the following observation is correct?



X

A

In test tube 'P' iron nail is coated with a blue coating and in test tube 'Q' there is no reaction.

B

Iron nail is coated with a brown coating in test tube 'P' and silver coating in test tube 'Q'

B

C

There is no reaction in either of the test tubes 'P' or 'Q'

D

There is no reaction in test tube 'P' but a silver coating on iron nail is seen in test tube 'Q'.

LET'S TEST YOUR MEMORY



Q. Which of the following reactions will NOT occur?

A



B



C



D

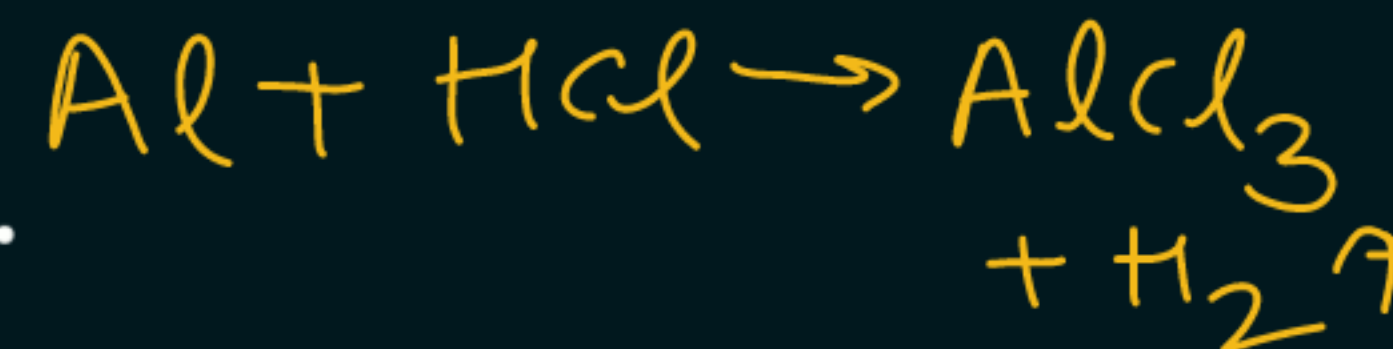


LET'S TEST YOUR MEMORY



Q. Batman has a utility belt made of aluminium (Al) on which he carries chemical weapons like concentrated hydrochloric acid (HCl) on special missions.

On a particularly long fight, he spills some of the acid on his belt.



Which of the following reactions does Batman observe? Note: Al is more reactive than H.

A

A



B



C

No reaction occurs.

D



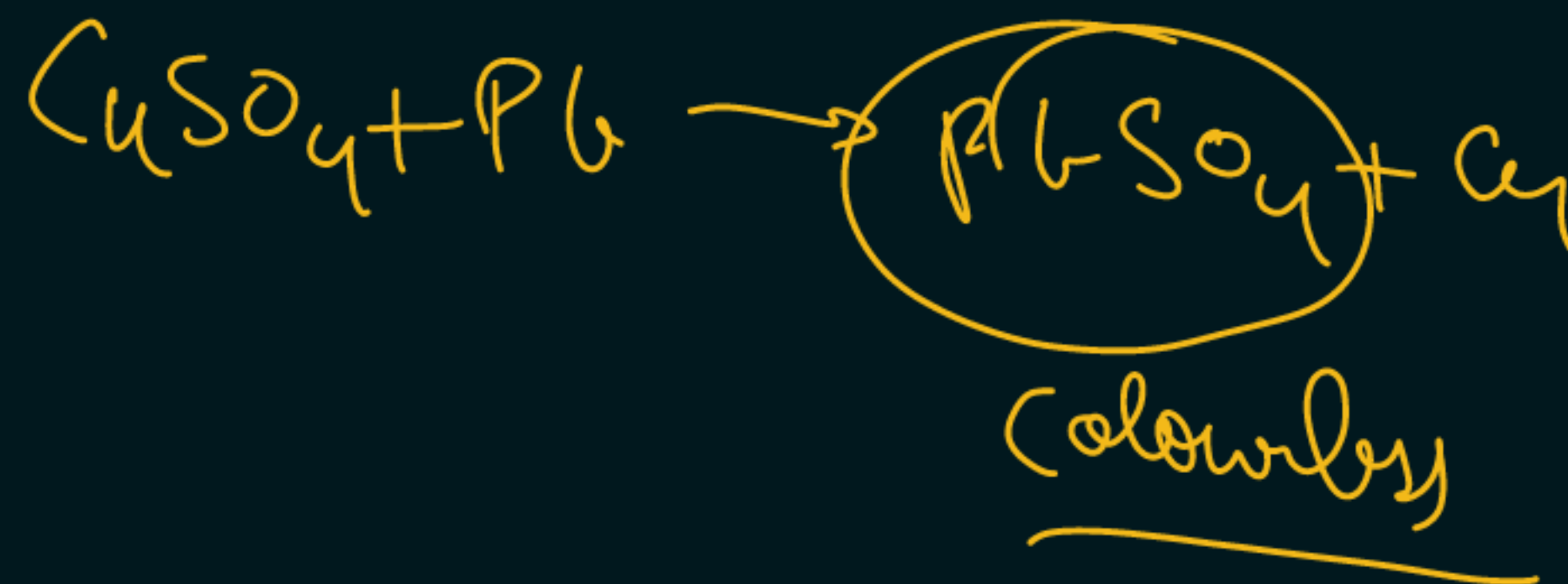
LET'S TEST YOUR MEMORY



Q. Assertion (A): The colour of aqueous copper sulphate solution turns colourless when a piece of lead is added to it.

Reason (R): Lead is more reactive than copper, and hence displaces copper from its salt solution.

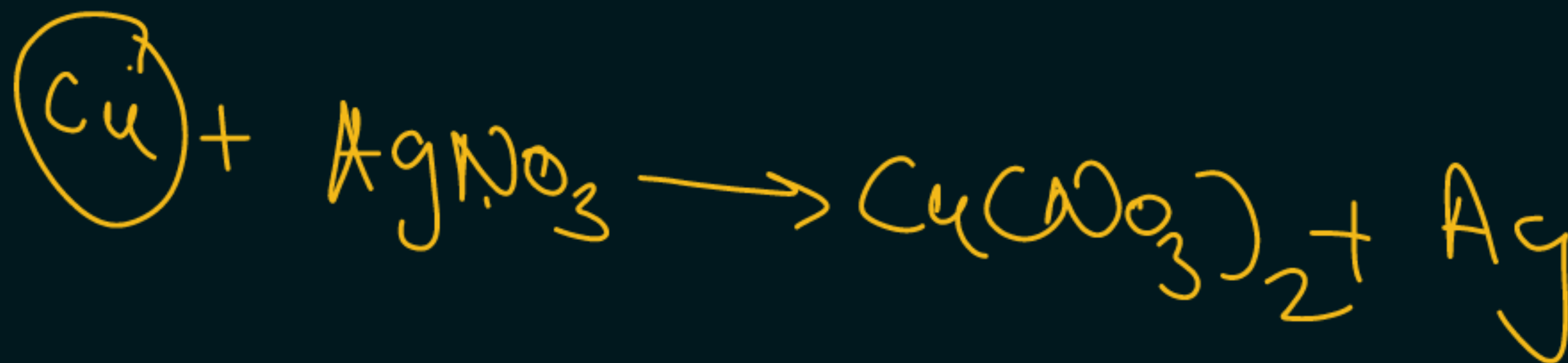
(A)



Q. In the refining of silver, the recovery of silver from silver nitrate solution involved displacement by copper metal. Write down the reaction involved.

In the refining of silver, copper displaces silver from silver nitrate solution because copper is more reactive than silver.

Chemical reaction: $\text{Cu(s)} + 2\text{AgNO}_3(\text{aq}) \rightarrow \text{Cu(NO}_3)_2(\text{aq}) + 2\text{Ag(s)}$



LET'S TEST YOUR MEMORY

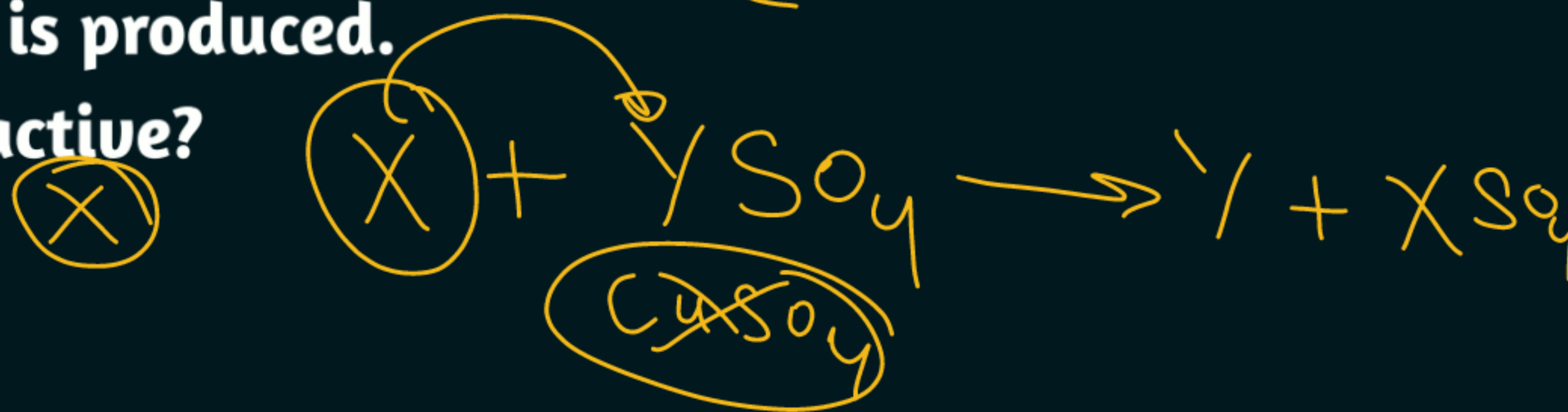


Q. A metal X is added to a solution of metal Y sulphate. A new metal Y is formed and X sulphate is produced.

(a) Which metal is more reactive?

(b) Identify reaction type

(c) Write general equation



Ans -

(a) Metal X is more reactive than metal Y.

(b) Displacement reaction (single replacement reaction).

(c) $X + YSO_4 \rightarrow XSO_4 + Y$

LET'S TEST YOUR MEMORY



Q. A metal X is dipped in a blue-coloured solution of salt Y. After some time:

- The blue colour of the solution fades.
- A brown layer forms on metal X.
- The reaction is:



(a) Identify X and Y.

(b) What is Z?

(c) Name the type of reaction.

(d) Give the reason for the colour change.



Ans -

(a) X = Iron (Fe), Y = Copper sulphate (CuSO_4)

(b) Z = Iron sulphate (FeSO_4)

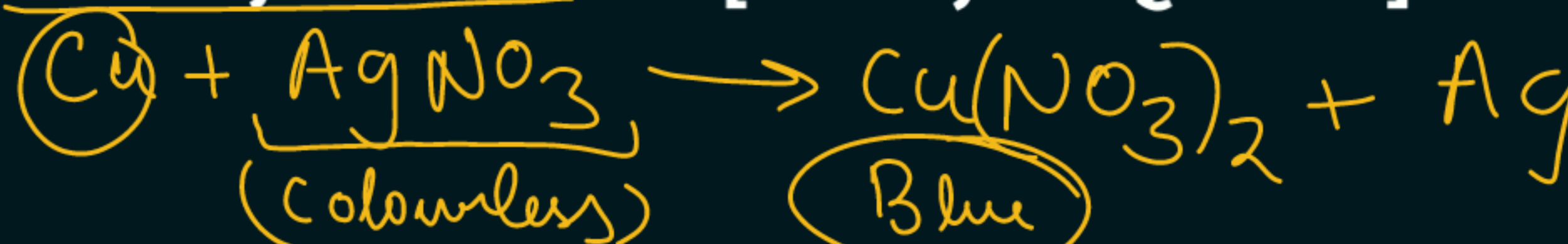
(c) Displacement reaction

(d) Iron displaces copper from CuSO_4 , so blue colour fades and brown copper deposits on iron.

LET'S TEST YOUR MEMORY



Q. A copper coin is kept immersed in a solution of silver nitrate for some time. What will happen to the coin and colour of the solution? [Delhi Gov. SQP 2025]



Ans -

When a copper coin is immersed in silver nitrate solution, a displacement reaction occurs.

Change in the coin: Silver metal gets deposited on the surface of copper coin, so the coin appears silvery.

Change in solution colour: The colourless silver nitrate solution turns blue due to the formation of copper(II) nitrate, which is blue in colour. This can be depicted by the following chemical reaction: $\text{Cu(s)} + 2\text{AgNO}_3(\text{aq}) \rightarrow \text{Cu(NO}_3)_2(\text{aq}) + 2\text{Ag(s)}$

LET'S TEST YOUR MEMORY



Q. A zinc plate was put into a solution of copper sulphate kept in a glass container. It was found that the blue colour of the solution gets fader and fader with passage of time. After few days, when the zinc plate was taken out of the solution, a number of holes were observed on it.

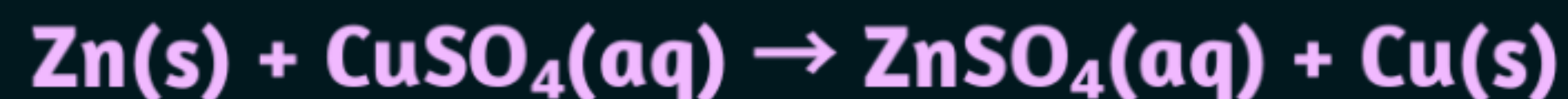
(A) State the reason for the changes observed on the zinc plate.

(B) Write the chemical equation for the reaction involved.

HW

(A) When a zinc plate is placed in copper sulphate solution, the blue colour fades because zinc is more reactive than copper and displaces copper from its solution. Zinc goes into the solution forming zinc sulphate, while copper gets deposited on the zinc plate. Due to continuous loss of zinc into the solution, holes are formed on the zinc plate.

(B) The chemical equation for the reaction is:



LET'S TEST YOUR MEMORY



Q. A student adds zinc granules to dilute sulphuric acid.

- (a) What observation will he notice?**
- (b) Name the gas evolved.**
- (c) Write the test for the gas.**
- (d) Write the balanced chemical equation for the reaction.**

HW

Answer: (a) Effervescence (bubbles) is observed due to the evolution of gas.

(b) Hydrogen gas (H_2) is evolved.

(c) The gas is tested by bringing a burning matchstick near the mouth of the test tube; it burns with a 'pop' sound, confirming hydrogen.

(d) $Zn(s) + H_2SO_4(aq) \rightarrow ZnSO_4(aq) + H_2(g) \uparrow$

LET'S TEST YOUR MEMORY



CBS
Q. During the time period of her experimental class, the science teacher Mrs. Priyanjali placed an experimental set-up in front of the students as shown in figure (a). It was observed that an iron nail is immersed in a solution of a compound 'P'. After a few hours, the students observed the change in the iron nail as shown in figure (b).

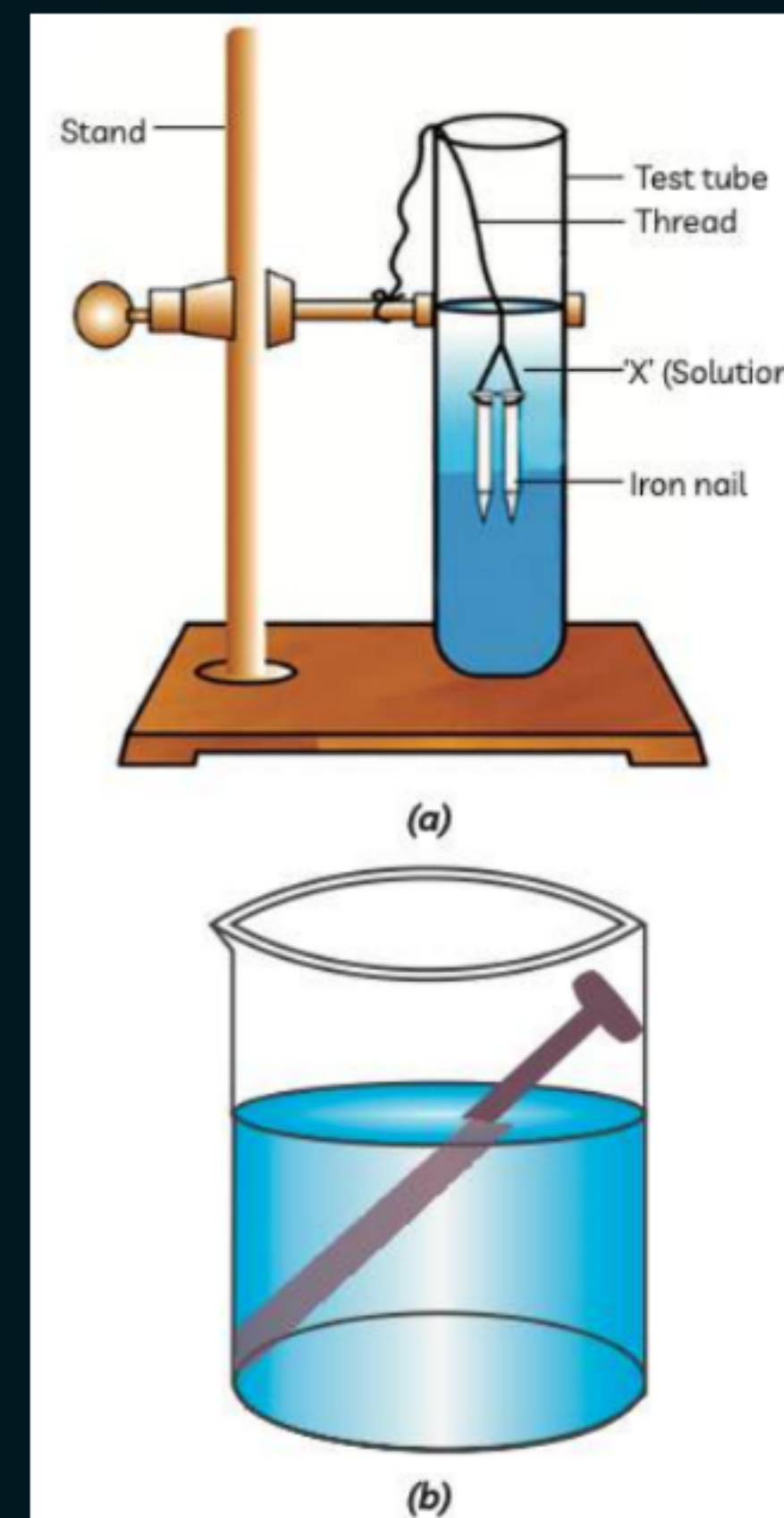
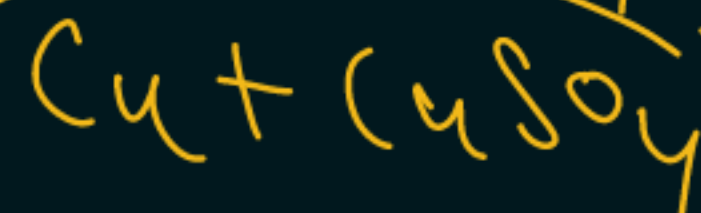
(A) Identify the chemical compound 'P'. What change did the students observe in the colour of solution of compound 'P'? Why did this change occur?

(B) Shreya, one of the students from the class, was trying to figure out the type of reaction that took place in the given set-up. Help her by identifying the same and give reason why is this reaction classified as such?

(C) After the experiment, the teacher told Shreya that she will demonstrate the same experiment next time with nails of two different metals. Will the same reaction take place if:

(i) Zinc nails are immersed in copper sulphate?

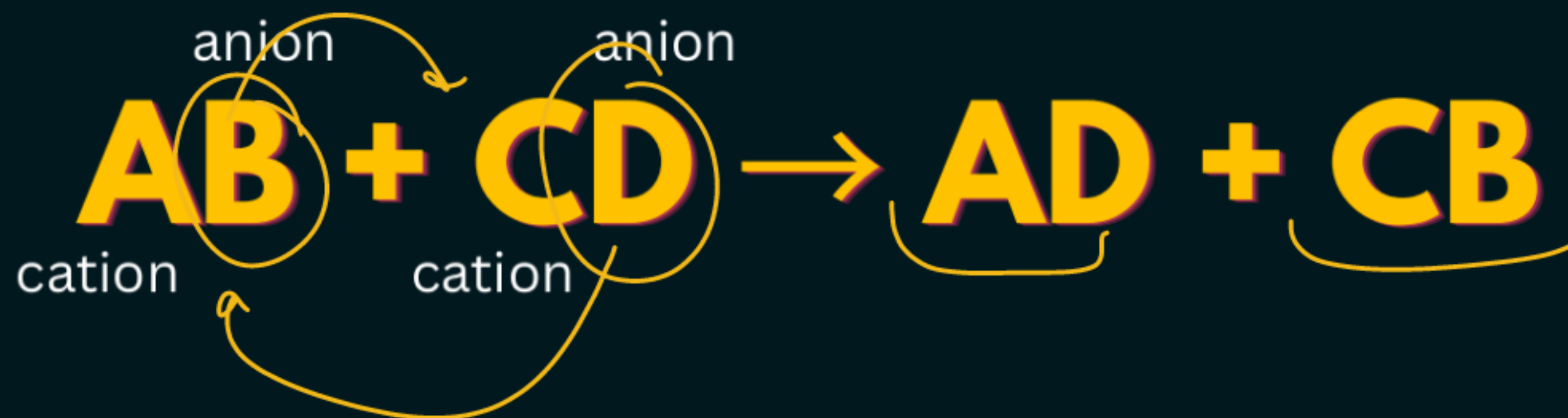
(ii) Copper nails are immersed in copper sulphate solution?



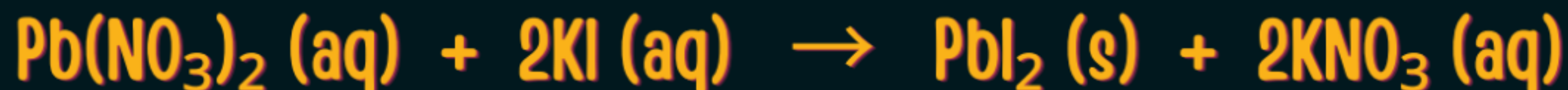
Double Displacement reaction

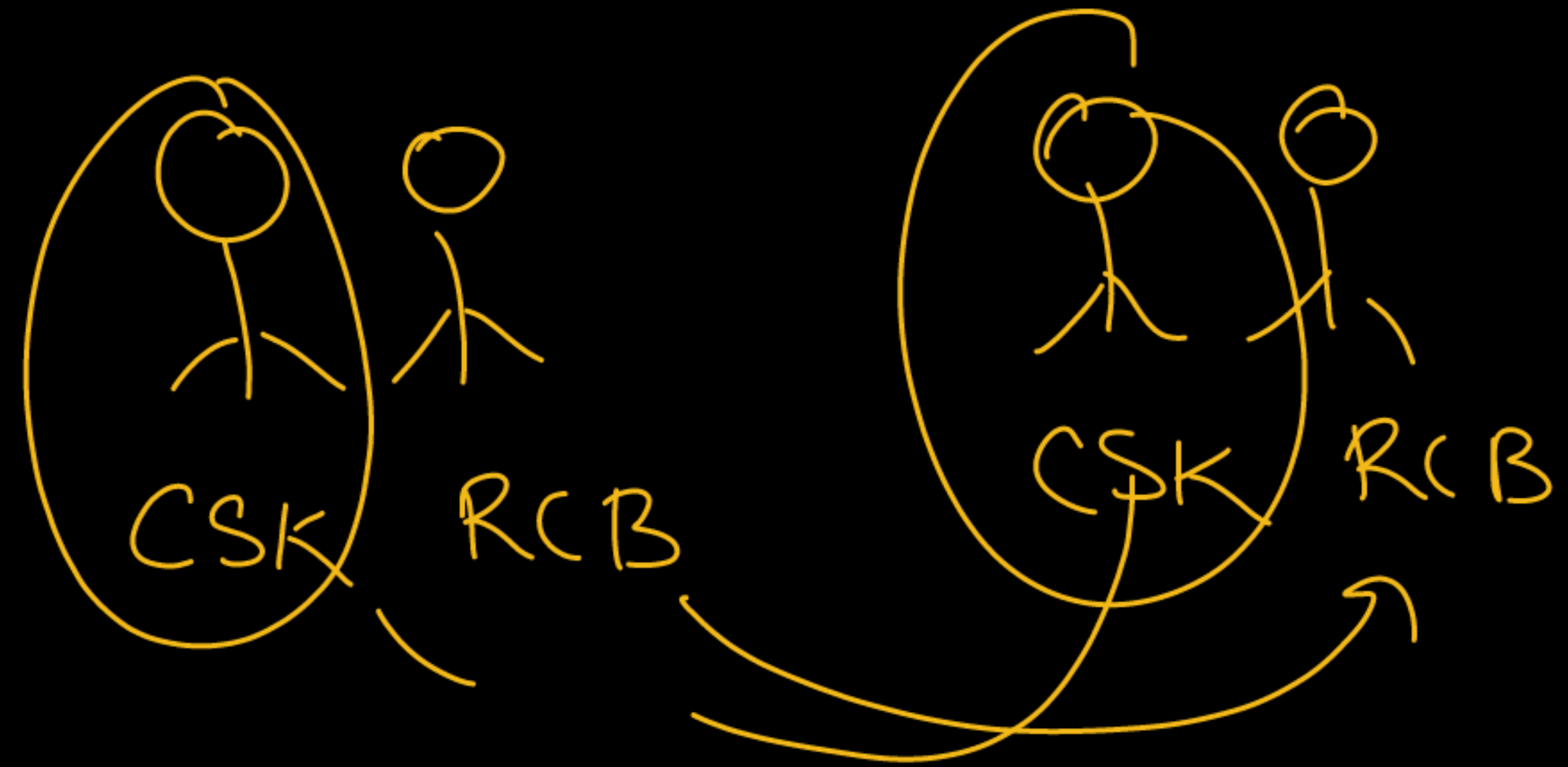
A chemical reaction in which the cations and anions of two different compounds exchange places to form two new compounds.

General
Equation



EXAMPLE





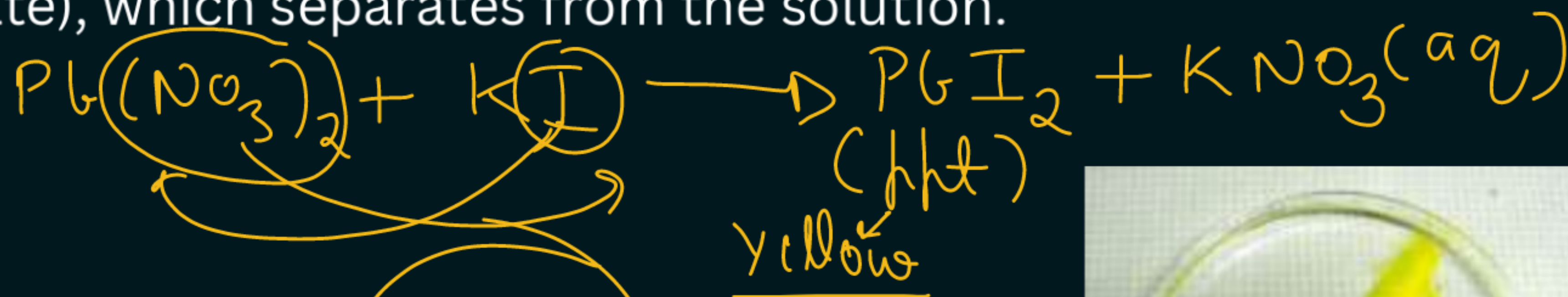
CSK CSK

RCB RCB

Precipitate Reaction

A reaction in which two aqueous ionic solutions react to form an insoluble solid (precipitate), which separates from the solution.

EXAMPLE



PK Points!



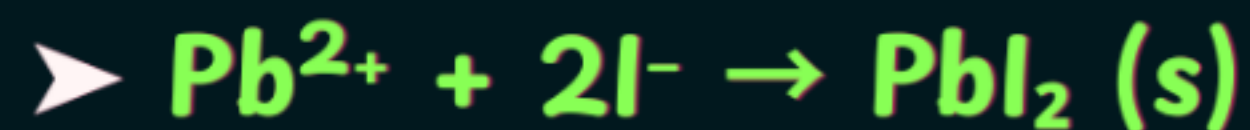
Observation: ➤ **Yellow precipitate forms** - Lead(II) iodide (PbI_2)

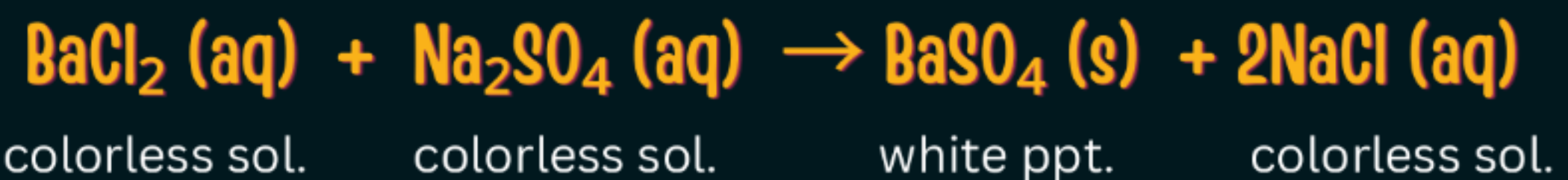
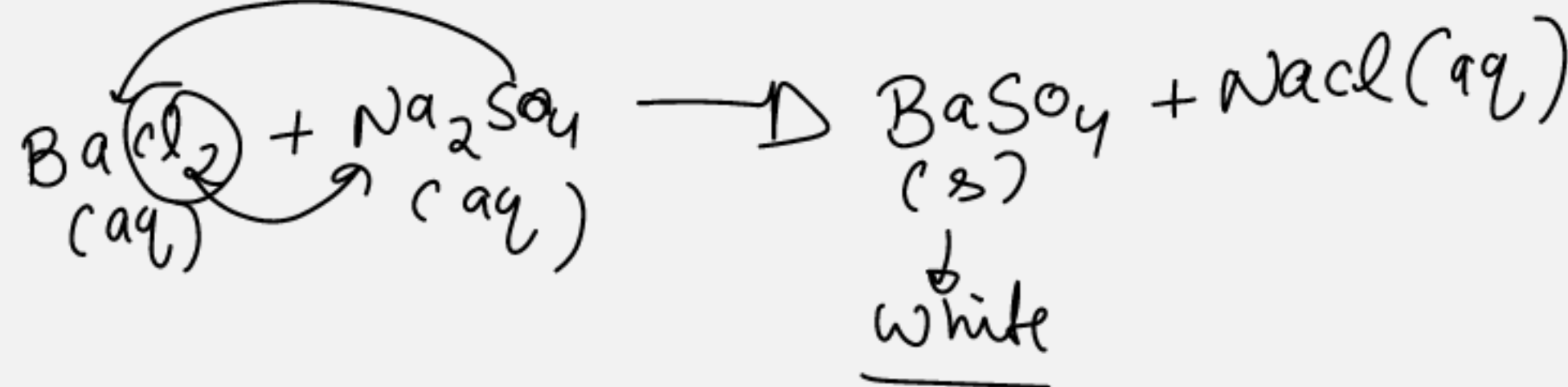
Balanced equation: ➤ $\text{Pb}(\text{NO}_3)_2 (\text{aq}) + 2\text{KI} (\text{aq}) \rightarrow \text{PbI}_2 (\text{s}) + 2\text{KNO}_3 (\text{aq})$

Type of reaction:

- **Double displacement**
- **Precipitation reaction**

Ions involved in precipitate formation:





Activity 1.10

- Take about 3 mL of sodium sulphate solution in a test tube.
- In another test tube, take about 3 mL of barium chloride solution.
- Mix the two solutions (Fig. 1.9).
- What do you observe?



NCERT ACTIVITY

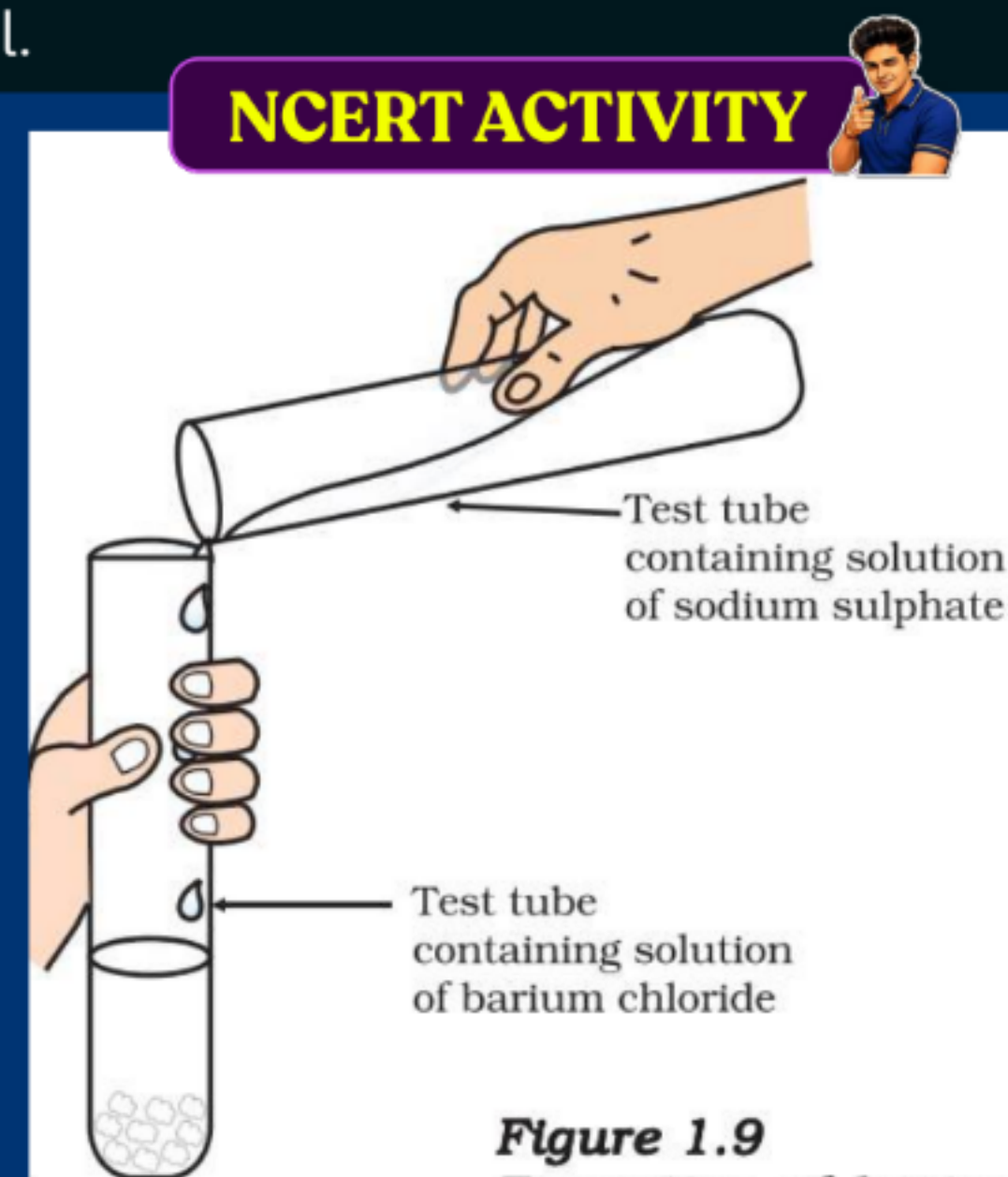
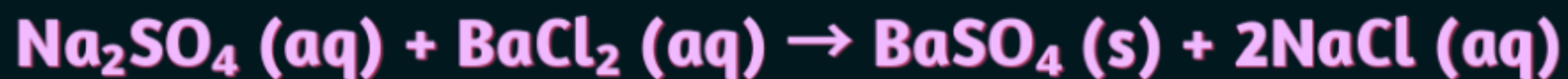


Figure 1.9
Formation of barium sulphate and sodium chloride

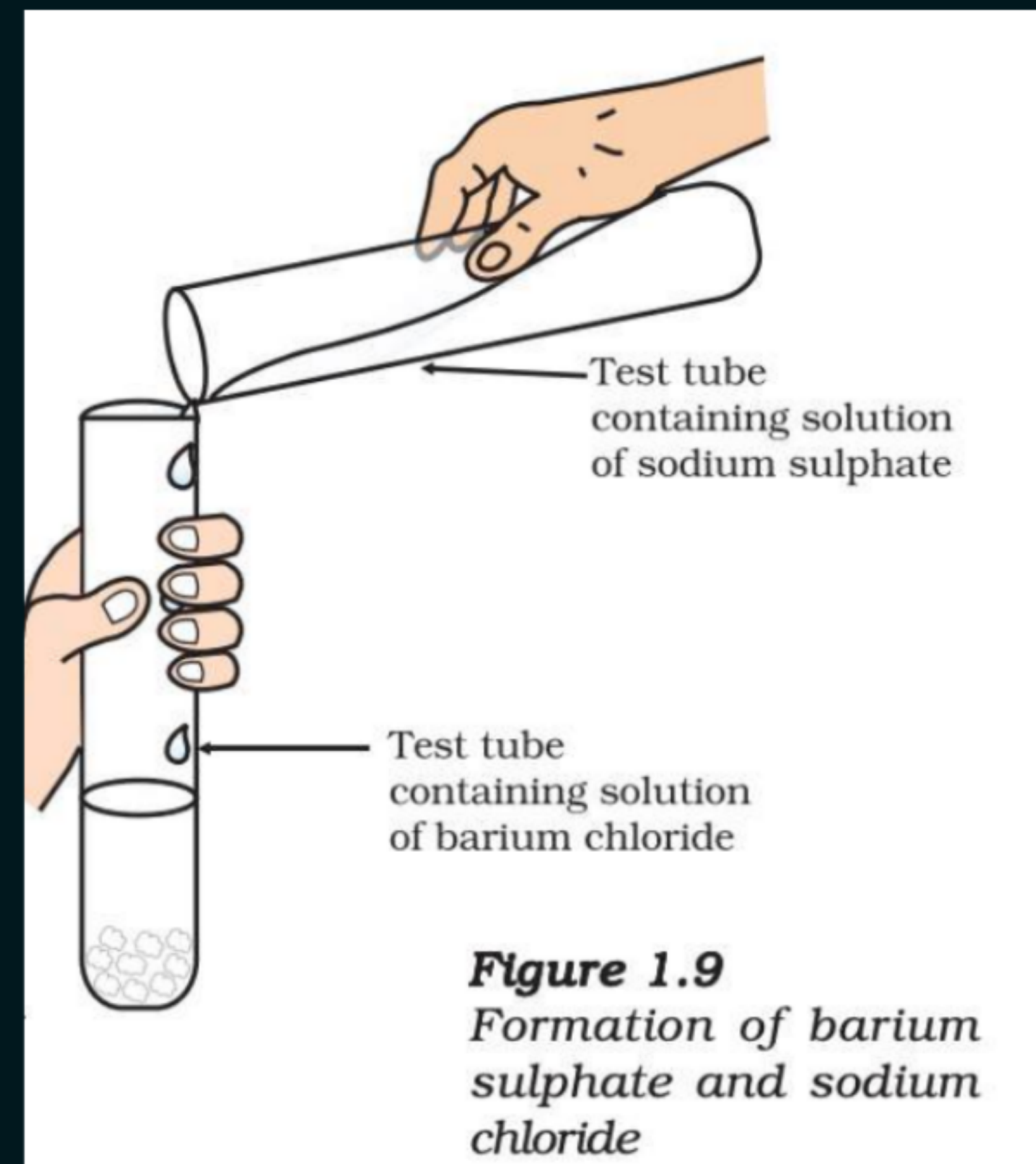
PK Points!



Chemical Reaction :



- BaSO_4 (barium sulphate) is a **white precipitate** and is insoluble in water.
- This is an example of a **precipitation reaction**.
- Also an example of a **double displacement reaction** (due to ion exchange).
- Ion exchange involved: Ba^{2+} reacts with SO_4^{2-}



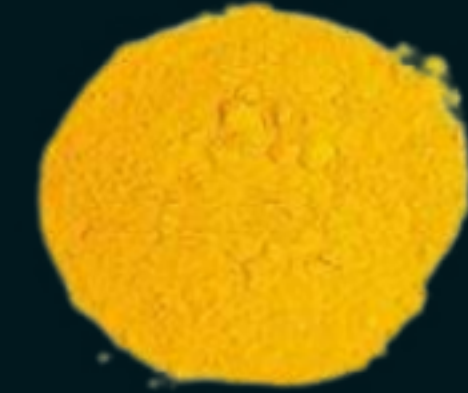
Precipitation Table Colour:

Precipitates	Colours
BaSO_4	White
PbI_2	Yellow
CaCO_3	White
AgCl	White
CuS	Black

BaSO_4



PbI_2



CaCO_3



AgCl



CuS



LET'S TEST YOUR MEMORY



Q. A student mixes two solutions but no precipitate, gas, or colour change is observed. What is the most likely conclusion?

B

A

A double displacement reaction has definitely occurred

B

No reaction has occurred

C

Only physical mixing has occurred

D

Reaction occurred but is not visible

LET'S TEST YOUR MEMORY



Q. Two aqueous solutions are mixed and a white insoluble solid is formed. What type of reaction has occurred?

P

A

Combination reaction

B

Displacement reaction

C

Decomposition reaction

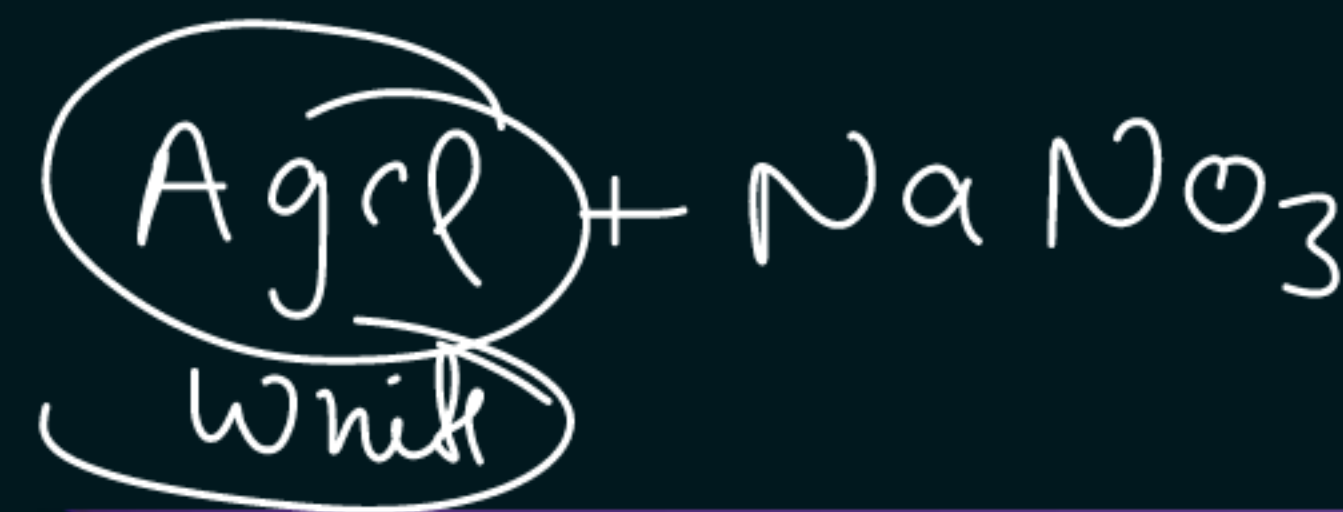
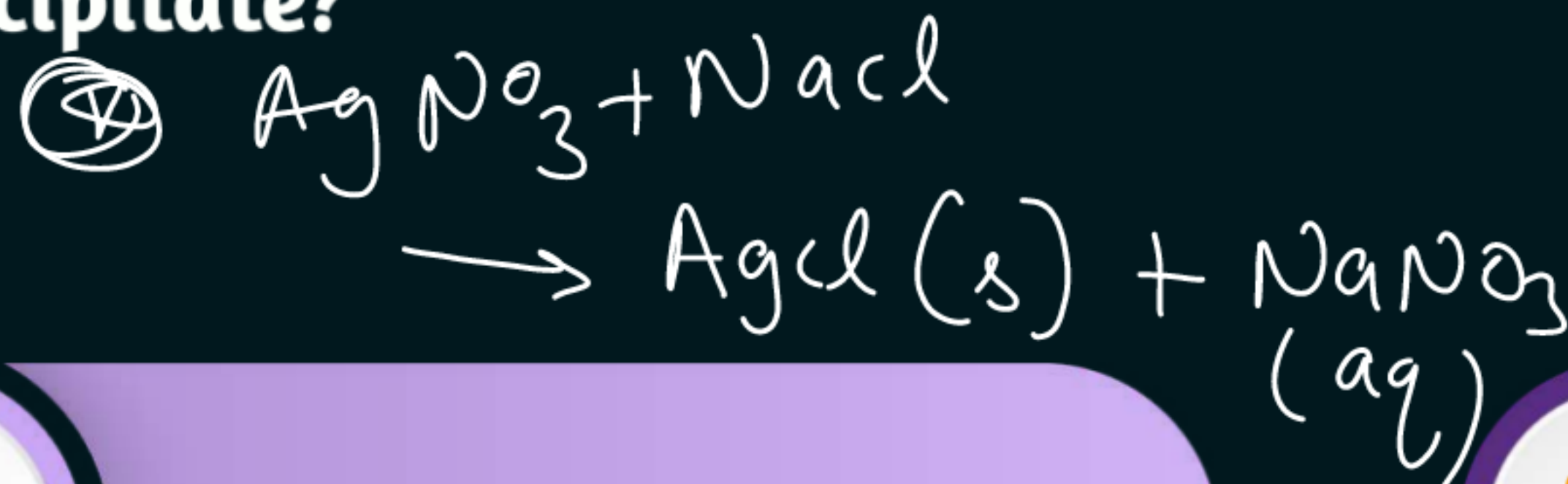
D

Double displacement (precipitation) reaction

LET'S TEST YOUR MEMORY



Q. Which of the following reactions results in formation of a precipitate?



A



B



C



D



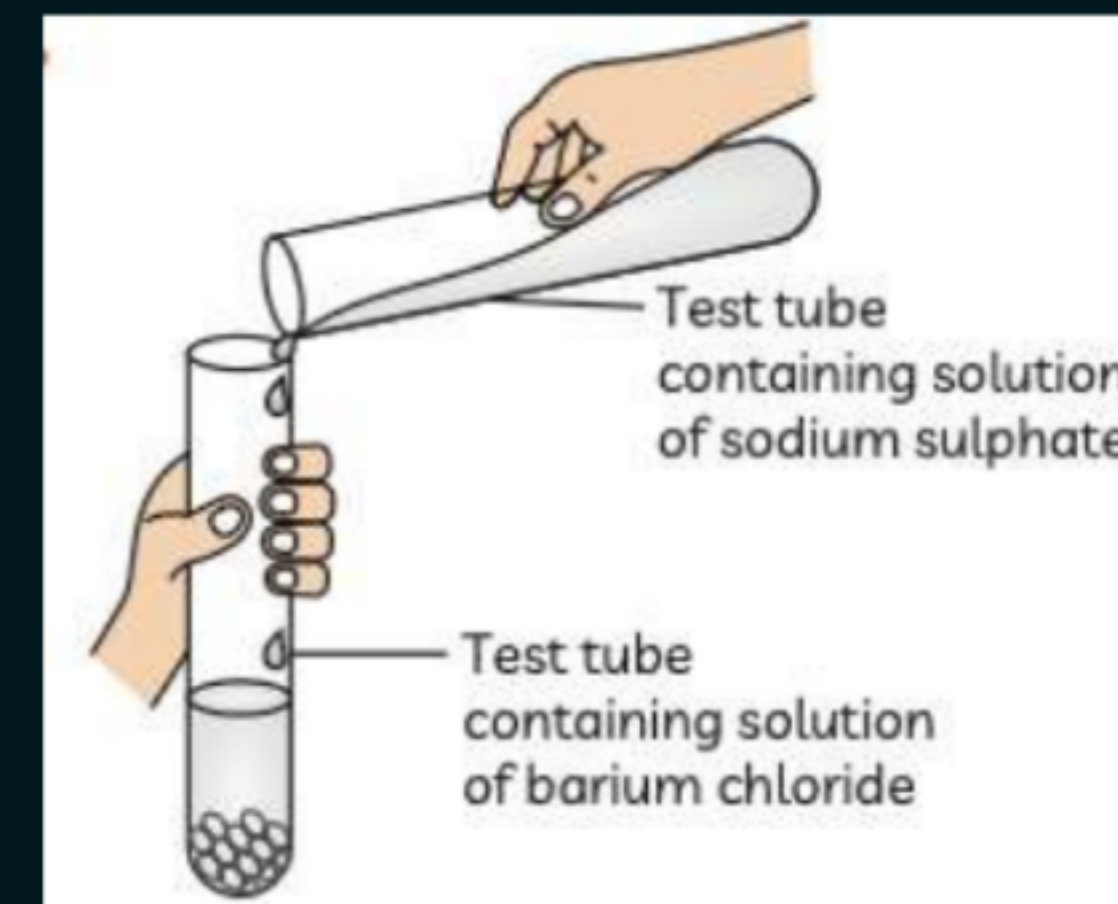
LET'S TEST YOUR MEMORY



आरंभ
BATCH

1/10

Q. A student performs an experiment by mixing two solutions. One test tube contains a solution of sodium sulphate, which is added to another test tube containing a solution of barium chloride. A white solid is formed in the reaction. Identify the product which represents the solid state in the above reaction.



A

Barium chloride

B

Barium sulphate

C

Sodium chloride

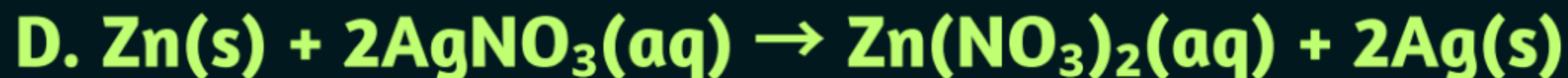
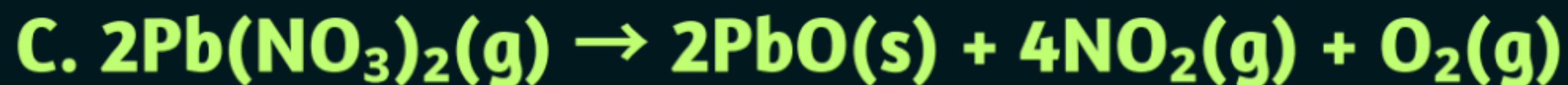
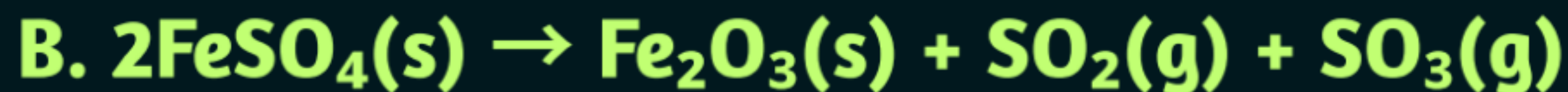
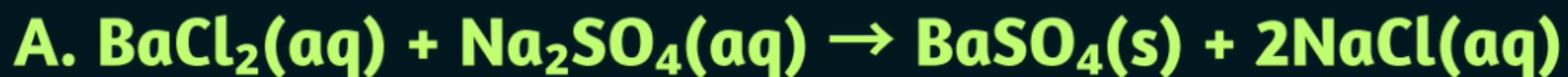
D

Sodium sulphate

LET'S TEST YOUR MEMORY



Q. Which of the following reactions represents a double displacement reaction?



HW